

Anti-proliferative activity and chemoprotective effects towards DNA oxidative damage of fresh and cooked Brassicaceae.

Epidemiological evidence shows that regular consumption of Brassicaceae is associated with a reduced risk of cancer and heart disease. Cruciferous species are usually processed before eating and the real impact of cooking practices on their bioactive properties is not fully understood. We have evaluated the effect of common cooking practices (boiling, microwaving, and steaming) on the biological activities of broccoli, cauliflower and Brussels sprouts. Anti-proliferative and chemoprotective effects towards DNA oxidative damage of fresh and cooked vegetable extracts were evaluated by 3-(4,5-dimethylthiazol-2-yl)-5-(3-carboxymethoxyphenyl)-2-(4-sulfophenyl)-2H-tetrazolium and Comet assays on HT-29 human colon carcinoma cells. The fresh vegetable extracts showed the highest anti-proliferative and antioxidant activities on HT-29 cells (broccoli>cauliflower = Brussels sprouts). No genotoxic activity was detected in any of the samples tested. The cooking methods that were applied influenced the anti-proliferative activity of Brassica extracts but did not alter considerably the antioxidant activity presented by the raw vegetables. Raw, microwaved, boiled (except broccoli) and steamed vegetable extracts, at different concentrations, presented a protective antioxidative action comparable with vitamin C (1 mM). These data provide new insight into the influence of domestic treatment on the quality of food, which could support the recent epidemiological studies suggesting that consumption of cruciferous vegetables, mainly cooked, may be related to a reduced risk of developing cancer.